INTELIGÊNCIA ARTIFICIAL

Aprendizado Descritivo
Clustering

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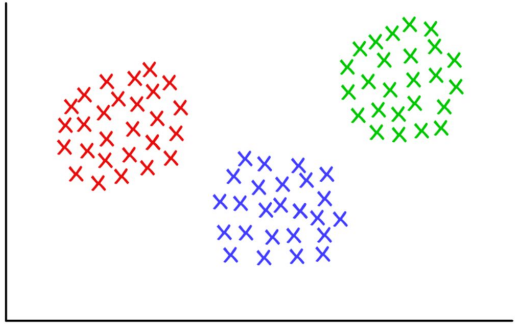
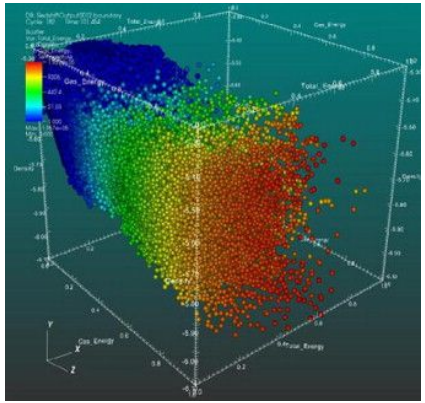
Curse of Dimensionality

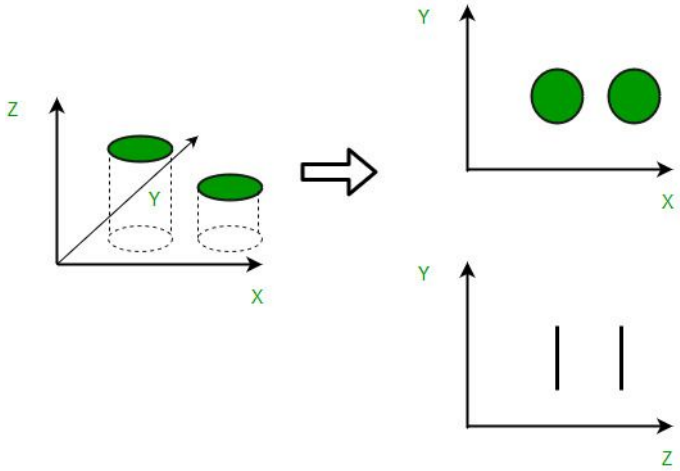
- Após aplicarmos uma medida de distância para todos os exemplos buscamos encontrar a **formação natural de grupos** (imagem da esquerda).
- No entanto, quando lidamos com um dataset de alta dimensionalidade pode ocorrer de não haver uma **separação clara entre os grupos** (imagem da direita).
- Para isso precisamos aplicar técnicas de redução de dimensionalidade antes de qualquer análise de grupos.

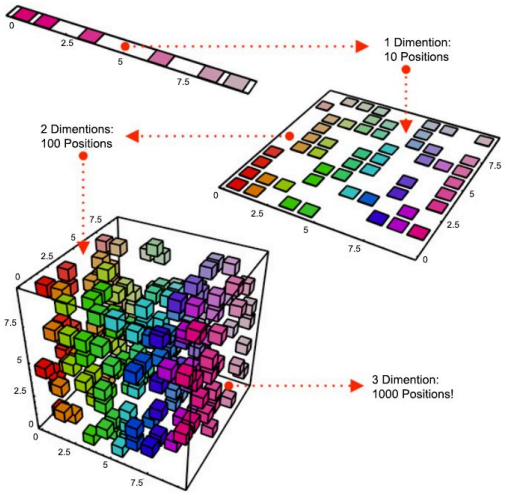




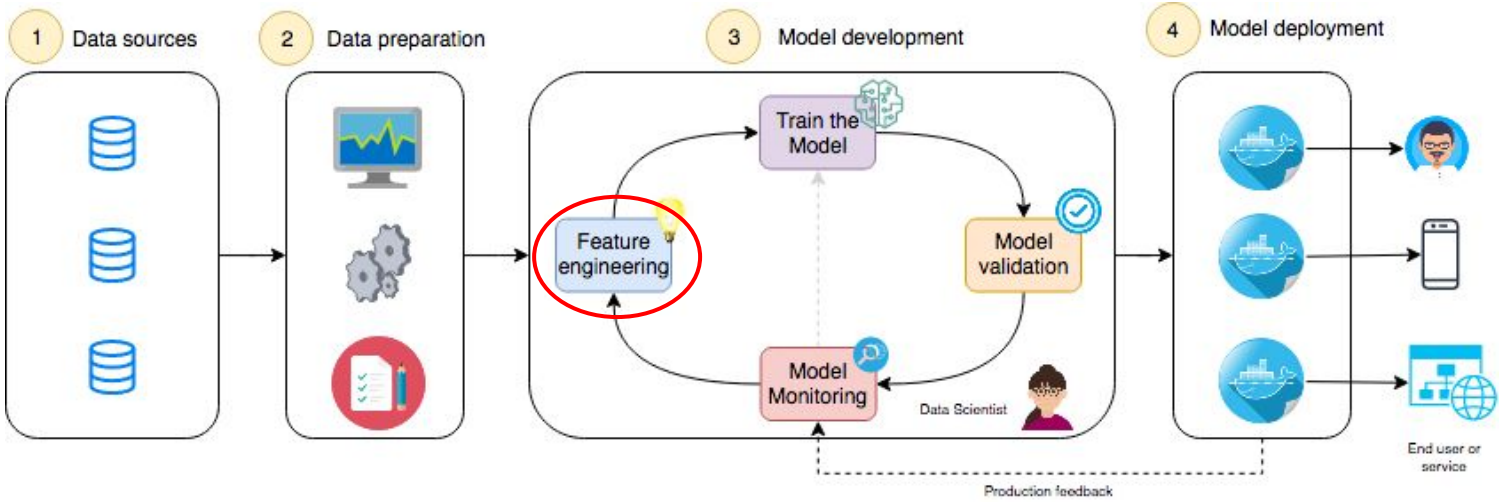
Curse of Dimensionality

- Além da dificuldade na **visualização dos grupos**, a alta dimensionalidade aumenta a **dispersão dos dados** e com isso prejudica a performance dos algoritmos
- Outro problema é o custo de armazenamento de uma quantidade excessiva de informação (exemplo, imagens de altíssima resolução)





Data Science Solution Workflow



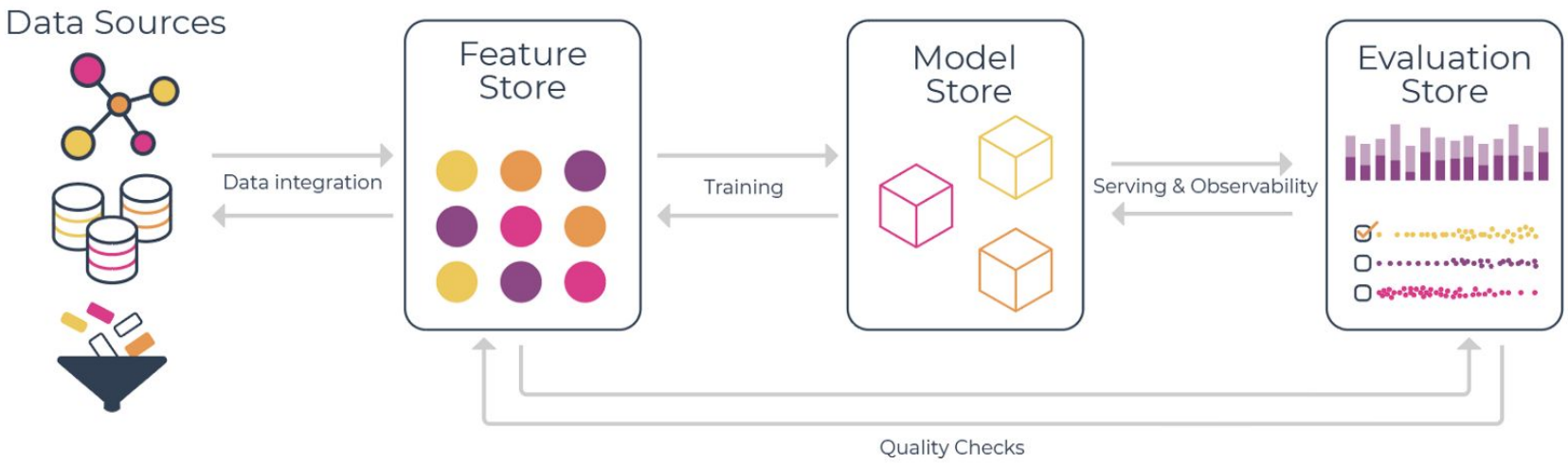

```python
# fit the model
model.fit(X_train, y_train)

# save the model
import pickle
pickle.dump(model, open("model.pkl", "wb"))

# load the model
model = pickle.load(open("model.pkl", "rb"))

# use model to predict
y_pred = model.predict(X_input)
```

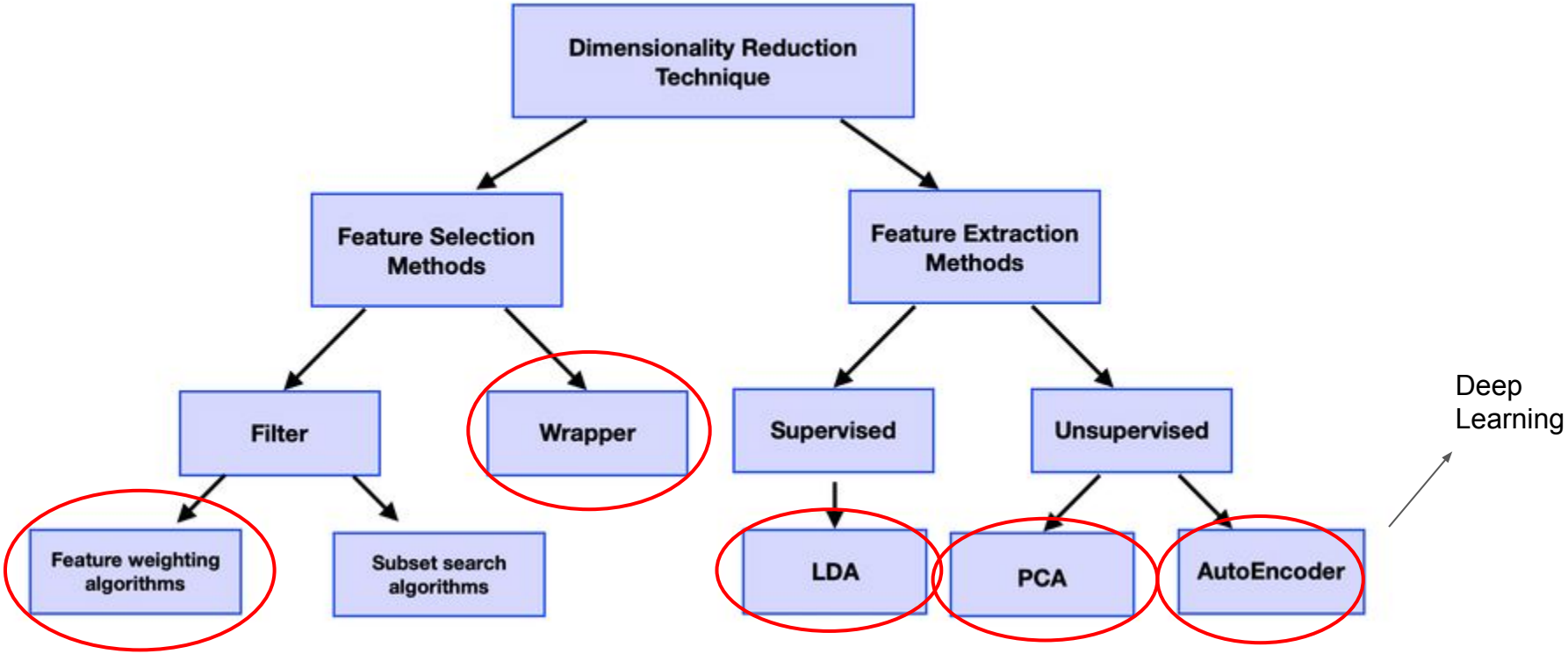
Data Science Solution Workflow



https://towardsdatascience.com/the-only-3-ml-tools-you-need-1aa750778d33

Redução da Dimensionalidade

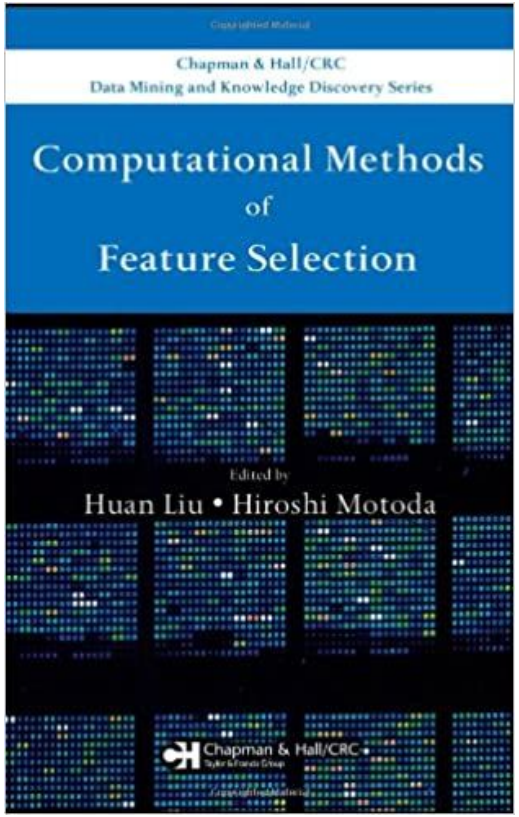
Uma visão **resumida** das técnicas de redução de dimensionalidade

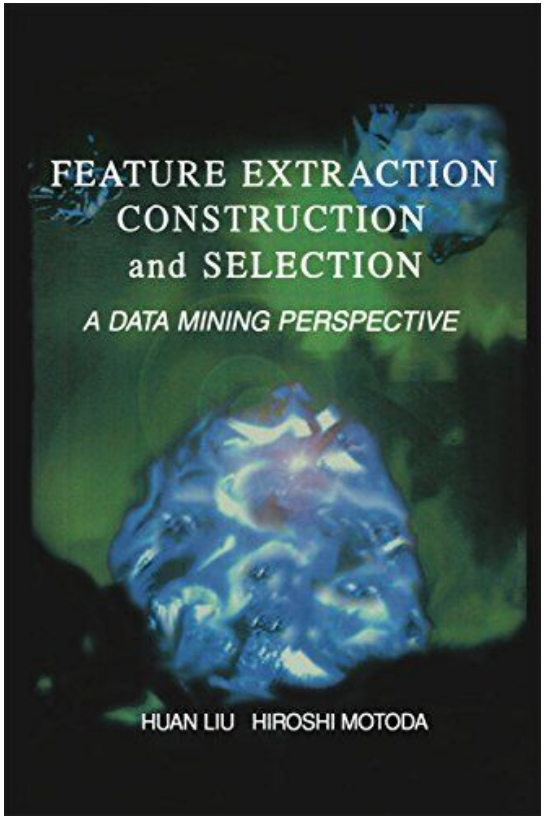


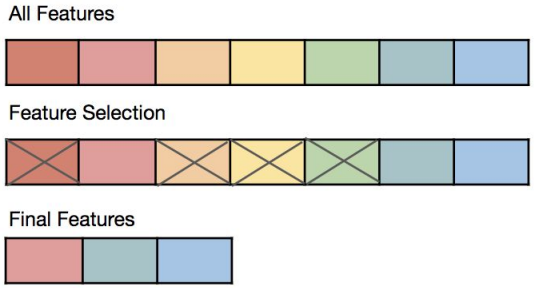
Redução da Dimensionalidade

Referências

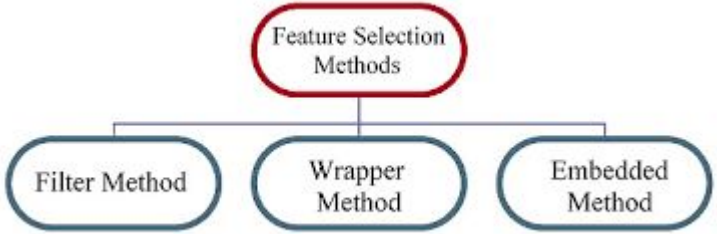
[preprint] Feature Selection: Data Perspective. 2018







1



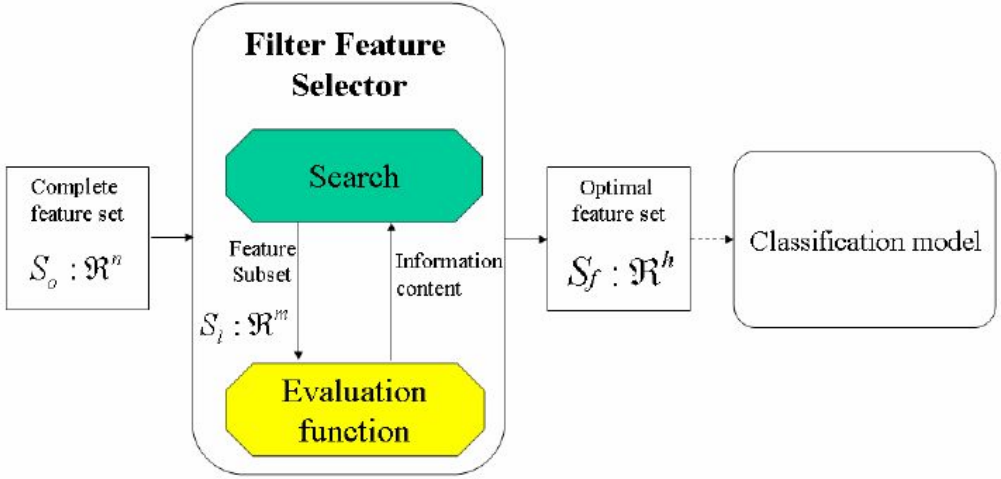
Seleção de Features

- Como podemos **selecionar** as features **mais relevantes** de forma **supervisionada** e **não supervisionada** ?

Filter Feature Selection

Filter Methods

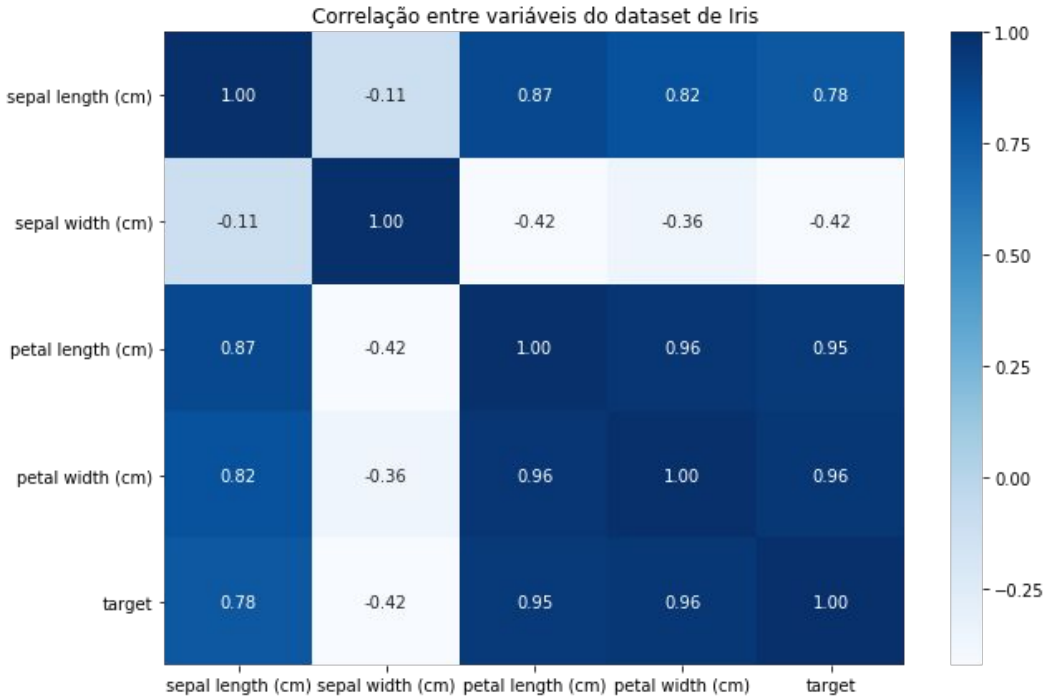
- Métodos do tipo Filter utilizam alguma propriedade dos dados para avaliação das features (ex.: filtro de variância)
- Quando supervisionado utilizam uma espécie de índice de correlação com a variável alvo
- Geralmente tendem a ser mais rápidos

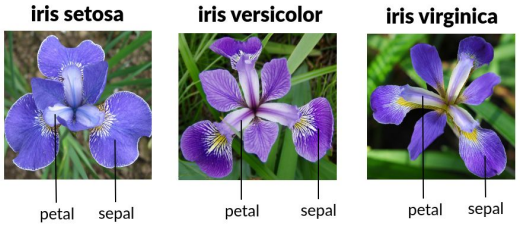


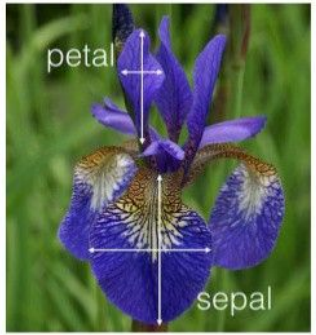
Filter Feature Selection

Análise de Correlação

$$r = \frac{\sum (x - m_x)(y - m_y)}{\sqrt{\sum (x - m_x)^2 \sum (y - m_y)^2}}$$







Training / test data

Features | Labels

Sepal length	Sepal width	Petal length	Petal width	Species
5.1	3.5	1.4	0.2	Iris setosa
4.9	3.0	1.4	0.2	Iris setosa
7.0	3.2	4.7	1.4	Iris versicolor
6.4	3.2	4.5	1.5	Iris versicolor
6.3	3.3	6.0	2.5	Iris virginica
5.8	3.3	6.0	2.5	Iris virginica

Filter Feature Selection

Análise de Correlação

Features				Labels
Sepal length	Sepal width	Petal length	Petal width	Species
5.1	3.5	1.4	0.2	Iris setosa
4.9	3.0	1.4	0.2	Iris setosa
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6.4	3.2	4.5	1.5	Iris versicolor
6.3	3.3	6.0	2.5	Iris virginica
5.8	3.3	6.0	2.5	Iris virginica

$$r = \frac{\sum (x - m_x)(y - m_y)}{\sqrt{\sum (x - m_x)^2 \sum (y - m_y)^2}}$$

x	y	x - mx	y - my	(x-mx)*(y-my)	(x - mx)²	(y - my)²
0.2	0	-1.18	-1.0	1.18	1.40	1.0
0.2	0	-1.18	-1.0	1.18	1.40	1.0
1.4	1	0.017	0.0	0.0	0.00	0.0
1.5	1	0.12	0.0	0.0	0.01	0.0
2.5	2	1.12	1.0	1.12	1.25	1.0
2.5	2	1.12	1.0	1.12	1.25	1.0
∑				4.6	5.31	4.0

= 4.6 / sqrt(5.31 * 4.0)
= 4.6 / sqrt(21.24)
= 4.6 / 4.61
= **0.99**

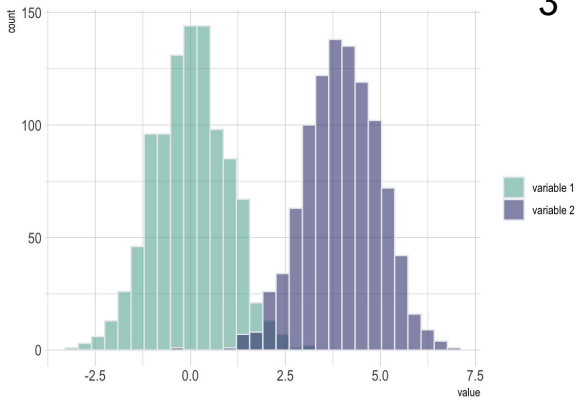
Filter Feature Selection

Análise da Variância e Histogramas

1

Offer	Age	Online payment	Items
1	35	0	5
1	36	0	4
1	41	0	9
1	34	0	10
1	38	0	3

3



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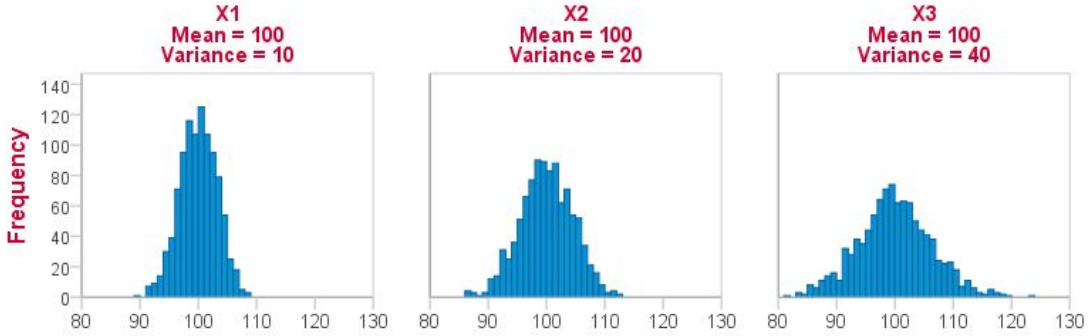
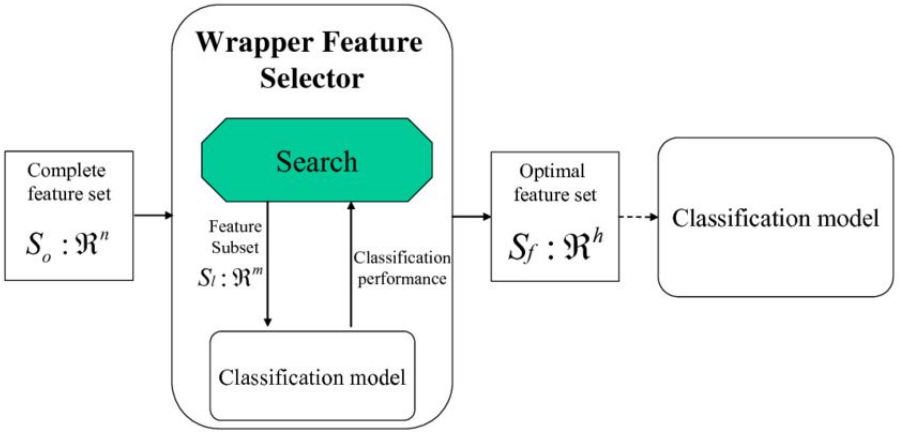


Wrapper Feature Selection

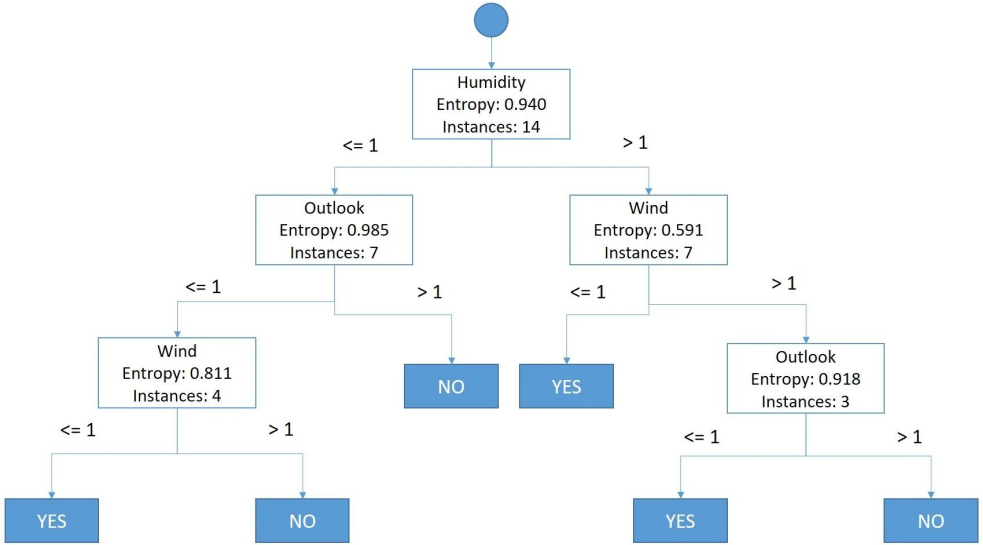
Wrapper Methods

- Métodos do tipo Wrapper fazem uso de um modelo preditivo para selecionar as features
- Quando não supervisionado acabam gerando pseudo-labels (clustering) para guiar a seleção (ex correlação ou informação mútua)
- Geralmente tendem a ser mais lentos devido ao passo de treinamento de um modelo



Wrapper Feature Selection

Árvores de Decisão



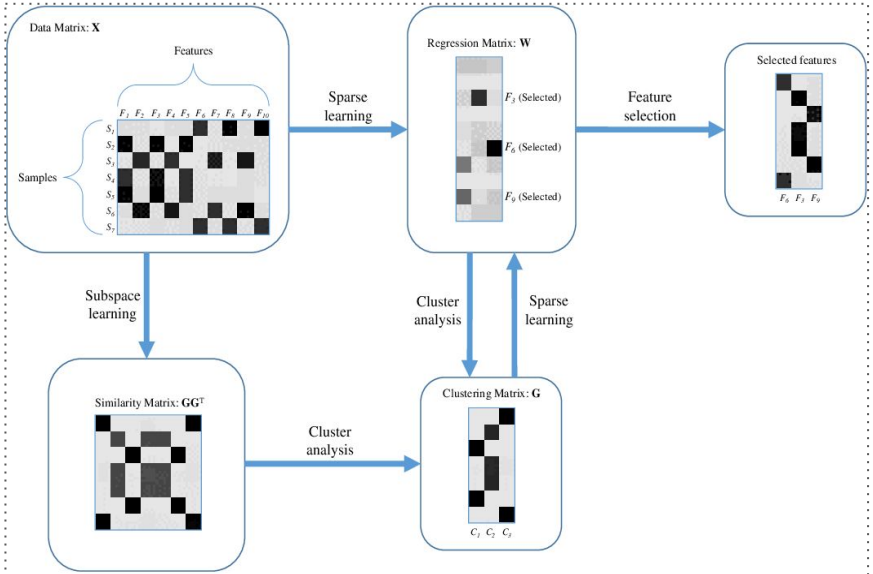
$$Gini = 1 - \sum_{i=1}^{n} p^2(c_i)$$

$$Entropy = \sum_{i=1}^{n} -p(c_i) log_2(p(c_i))$$

where $p(c_i)$ is the probability/percentage of class c_i in a node.

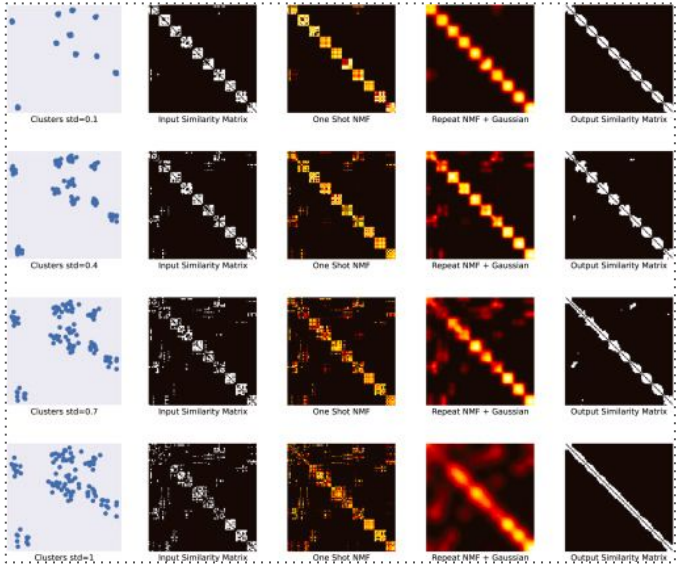
Wrapper Feature Selection

Subspace Learning



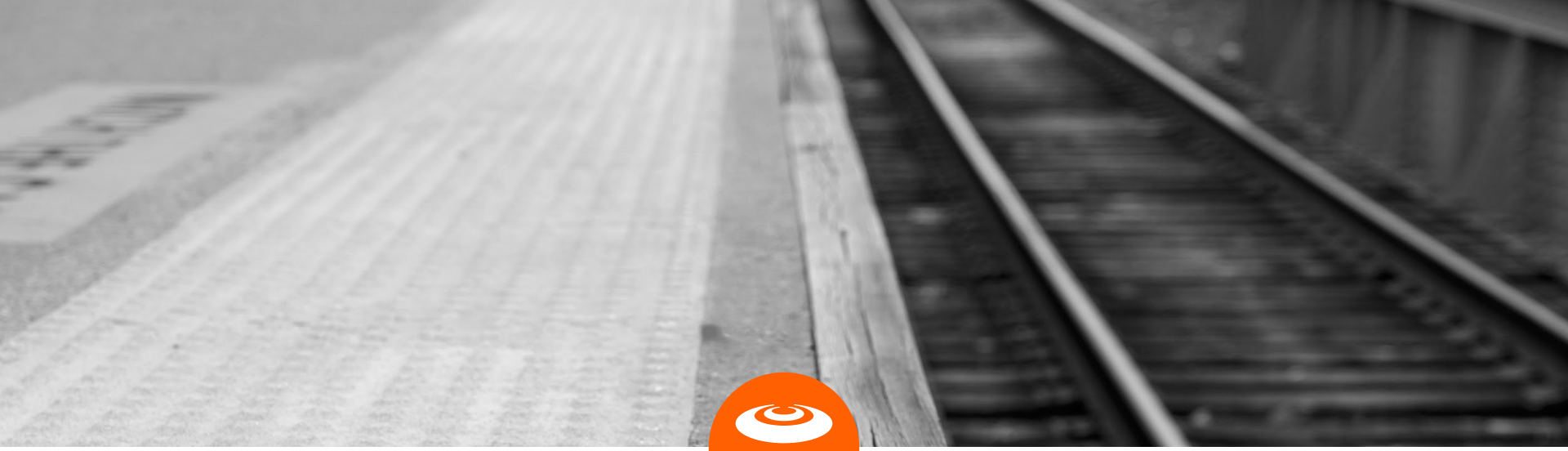
Parsa, M. G., Zare, H., & Ghatee, M. (2020). Unsupervised feature selection based on adaptive similarity learning and subspace clustering. *Engineering Applications of Artificial Intelligence*, *95*, 103855.

github: https://github.com/mohsengh/SCFS



Oliveira, M. D. S., Queiroz, S. R. D. M., & de Carvalho, F. D. A. (2022). Unsupervised feature selection method based on iterative similarity graph factorization and clustering by modularity. *Expert Systems with Applications*, *208*, 118092.

github: https://github.com/marcosd3souza/KNMFS



Hands On!

Experimentos em Machine Learning com Python

C . e . S . A . R

Pessoas impulsionando inovação.

Inovação impulsionando negócios.

NOSSO CONTATO

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